STRICTLY CONFIDENTIAL

Ornithoptera alexandrae: current status on Managalas Plateau and at Higaturu estate

Itinerary:	KL-Port Moressby	Mon. 18 Sept.
	Port Moresby-Popondetta-Managalas Plateau	Tues 19 Sept.
	Managalas Plateau	Tues. 19 Sept.
	Managalas Plateau	Wed. 20 Sept.
	Higaturu Estate	Thur. 21 Sept.
	Higaturu Estate-KL	Sat. 23 Sept.

Background

In the light of Sime Darby Foundation's generous grant to set up and staff for 3 years a laboratory at Higaturu Estate, it was agreed desirable to check out the status of Queen Alexandrae Birdwing *(Ornithoptera alexandrae)* ('The Birdwing') on the Plateau as well as the status of the butterfly on the Higaturu Estate Lejo Forest Reserve. It was believed the Plateau might be providing the remaining reserve for the last viable population of the butterfly.

I was accompanied on the visit by Charles Harbottle, an experienced entomologist who had previously spent some time investigating the status of the butterfly in and around Higaturu Estate.

The Managalas Plateau

The Plateau lies to the S.E. of Popondetta, and is accessed by a road turning off the road to the sea, about 2 miles out of Popondetta. An excellent tarmac surface gives way after about 3 miles to a single track and at times deeply rutted route, initially over low coastal hills covered with a native Acacia, and open lallang beneath – apparently fire climax vegetation. The road then rises to the plateau – about 600 – 700 m above sea level. The road was originally constructed around 1976 and for some years supported a bus service. The poor and deteriorating condition of the road makes this no longer feasible and the road is only open in dry weather to 4 w-d jeeps. We turned off just before dusk to the hamlet of Dareki – 7A Ward 3 – one of the 18 such on the Plateau, extending to 360,000 ha. Of this 80,475 ha is conserved. The formalization of the conservation area is to be celebrated in the next couple of months. The threats from mineral exploitation of the area (silver, nickel etc.) are thus reduced.

As a result of appalling roads the population of the Plateau, which has risen in recent years to about 22,000 persons is poorly served: health and education facilities are minimal, there is no power, and subsistence agriculture is the norm: bananas, taro, coffee and a little cocoa, the latter greatly reduced by the activities of the cocoa pod borer. If roads were to be improved, there are likely to be markets on the coast for agriculture products. This of course would put further pressure on the remaining forest areas where the Birdwing occurs.

NGO Involvement

There are a number of NGOs claiming involvement in the area.

1. Partners with Melanesians. Kenn Mondiai who kindly accompanied us on our visit is one of the leading figures in this NGO. Kenn is extremely knowledgeable, way beyond his formal training as a forester. He has been involved, after a stint at the University of Papua, since 1989. This is a national NGO, registered on 8.8.98. All members of its Board are PNG citizens. The NGO is entirely dependent on donor funding. They have partnered with the Rainforest Foundation of Norway since 1996. They strongly support the birdwing conservation programme. (But see his response to our press release, attached as an Appendix.)

2. The Managalas Conservation Foundation (MCF). This is an umbrella organisaton for all community based organizations on the Plateau.

3. Dadakul (Butterfly) a local organization works with MCF.

As far as we could gather, these organizations rub along together. It will be important for SDF to be aware of, but not get involved in the politics of these organizations.

There was some dissatisfaction with Ausaid's performance. When they left, they divided their assets between different local organisations and nobody really benefitted. The operation run by Cornwell Nukara and his brother Solomon Kotumba Bogina is a private initiative which survived the pull out of Ausaid.

Existing Birdwing Breeding Operations on the Managalas Plateau.

The only existing Birdwing breeding operation on the Plateau is that of Cornwell Nukara and his elder brother mentioned above. It operates about 40 minutes' walk from Dareki village – the walk involves wading through a couple of rivers and on through slightly overgrown vegetable gardens, and leads to a small hamlet of 2 or 3 houses on the edge of the forest. Inside the forest, much of the undergrowth has been cleared, except for saplings bearing *Aristochia dielsiana* creepers. Many of these carried larvae or pupae of *O. alexandrae* or *O. priamus.* The larvae appear to feed by preference on vines about 2-6 m above the ground.

When Cornwell and his colleagues find 6th instar larvae only, they transfer them on a length of cut foodplant to the 'greenhouse' a wooden framed building about 4 m x 17m, 2.5 m high, 3 m at the apex, Here the larvae are able to pupate in a protected environment, and emerge after 8 weeks as pupa (6 weeks on the plains) to extend and dry their wings before they are released back into the forest.

In answer to our query, the brothers indicated that although they located and monitored all smaller larvae within a radius of about 2 miles, they only collected the final instar larvae. They said they could not afford to collect the younger larvae in view of the work involved.

When we asked to see some of the younger larvae which were not being harvested till the 6th instar, we were met with excuses and prevarication – difficult to get through the forest etc. We, naturally, saw no sign of any commercial business in the insect. However I was informed in

Port Moresby that it is most unusual for PNG smallholders to maintain a project, apparently unremunerated, for so long. Normally dead specimens are smuggled out across the Indonesian border into W. Papua, and from there to Singapore, where they are imported as Indonesian material. From there they are retailed on the black market to collectors in the developed world. We should be more aware of, and keep our eyes open, to this possibility. We should bear in mind that initially at least we may be dependent on the goodwill of Cornwell & family to obtain breeding stock of *O. alexandrae* to get the laboratory started, and they may not be best pleased if as a result of our breeding operations, the black market price drops – and with it the pressure on wild populations. At the same time we do not wish to be associated with any aspect of the black market operations. It is a matter which may well require some careful thought.

On the basis that each week they are rearing 1 or 2 *O. alexandrae*, this suggests, that with no seasonality, as reported, they are breeding about 100 specimens a year: a significant contribution to the population with a radius of about 2 miles. It is not clear what levels of population exist elsewhere on the Plateau, but they may be not dissimilar.

We considered how best this population could be monitored, and concluded that probably a tally of the numbers hatched each week would be the most reliable census. The balance between ♂ & ♀ was reported to be close to 50 : 50, and disappearance/loss of *O. alexandrae* on the vines was confined to the early instars.

Status of *O. alexandrae* at Higaturu Estate.

About 3 years ago Charles Harbottle spent some time based at Higaturu Estate, with a view to establishing the status of *O. alexandrae* at and around Higaturu Estate. In particular he had retained records of the precise locations where 3 larvae had been found in 2014 on Aristolochia vines on the estate's Lejo Forest Reserve – a reserve of just under 50 ha. We revisited the site and were delighted to discover that on precisely the same spot there was one half grown *O. alexandrae* larva, which had been discovered about 3 days before our visit. Other larvae and a pupa (suspected male) have since been found at the same spot. So we can confirm that there is still a small residual population on the estate.

Unfortunately the reserve has been subject to much human disturbance, particularly near the Aristolochia vines.

However in a way more interesting was the foodplant it was on. This was an Aristolochia vine completely different from the one on which we had observed 6th instar larvae feeding on the Managalas Plateau. Proof of this was the photograph taken 2 days before by the estate assistant manager, Kenneth. This shows a fully opened flower, significantly different from the flowers of what we assume was *A. dielsiana* on the Managalas Plateau. The flower had fallen and disappeared by the time we arrived at the site, but we have asked Kenneth to forward the photograph.

This may provide a clue to the riddle of the foodplant, where it has been reported that some plants can provide food for the larva, while others prove toxic. It is just possible that there are 2 discrete populations, one feeding on the previously unknown vine in and around Higaturu and another feeding on *A. dielsiana* on the Mangalas Plateau. Whether these are subspecies, or conceivably a completely separate 'crypto' species, only DNA analysis will reveal. But is is

certainly a very exciting prospect.

We believe that there may also be one or two remaining smallholder areas around Higaturu with similar relict populations of the lowland form of the species.

At all events it will be important to collect cuttings from the vine at the Lejo FR, mark them separately and grow them on in the Higaturu nursery.

The Proposed Lab & Nursery at Higaturu.

While on the estate we were shown the proposed site for the entomologist's house, in the housing compound, and also the secure area adjacent to Sangara Mill where it is proposed to build the laboratory and housing for the guards. For security purposes we felt the guards' houses should be completely fenced off from the lab. We also thought it might be best to move the lab a few feet closer to the back fence so that it could benefit from afternoon shade provided by the adjacent mature palms.

At the nursery we were shown cuttings of *A. dielsiana* sent down by Cornwell from Managalas. There were approx. 200 of these, planted in polybags under shade. A number of them have already begun to shoot. It was suggested that in future the corky stem be nicked with a penknife to assist rooting and the stem laid horizontally and covered with +/- 2 cms of moist friable topsoil.

Back in Port Moresby, Ian Orrell pointed out that for 'speed cuttings' it may be best to tie 5 or 6 polybags together, and as soon as the shoot from the central bag is long enough, peg it down in an adjacent bag to develop its own roots This has been done successfully at WNB and we should probably adopt it at Higaturu.

There are already a few *A. tagala* vines in polybags at Higaturu, and it would be a good idea to let Cornwell and his colleagues have 2 or 3 of these to plant on his own land (where the plant is rare or non-existent: see next section.)

In future individual vines from which cuttings are taken both at Managalas and Lejo should be numbered and marked, the relevant cuttings likewise so that we can trace each plant back to the parent.

The best way of growing the vines, once they have struck root, is probably to let them grow over a trellis of rosewood as is being done in the nursery-making sure that neither the rosewood nor the vines grow higher than 6' to ensure ease of harvesting the leaves. Rosewood apparently grows well from cuttings.

The aim should be to have adequate *A. dielsiana* vines, both in the compound round the laboratory, with a plentiful back-up supply in the nursery in case of sabotage.

Next Steps

The SDF brief requires us to interact with and encourage smallholders to become involved in breeding work. It is clear, subject to the reservations expressed above, that the prime

candidates are Cornwell & his family. Probably the best approach would be:

(i) To supply him with some polybag seedlings of *A. tagala.* This is the foodplant of *O. goliath,* another very colorful species recorded on Managalas Plateau. However *A. tagala* only occurs rarely there – and Cornwell's plot apparently does not have specimens. If wild *O. goliath* could be encouraged in Cornwell's plot, he would be in the remarkable position of being able to show eco-tourists 3 Ornithoptera species, which should be very beneficial for him.

(ii) His 'brother'/cousin Solomon Kotumba Bogina has attended ½ an agricultural course at one of the PNG colleges, and was obliged to leave due to lack of funds. It might be a good idea to finance him to do the remainder of his course (1 – 1.5 years?)

(iii) At a later stage, if all goes well, we might consider, as has been done at the Dareki village, providing materials for a basic rest house/hostel at Cornwell's plot of land. That at Dareki has 5 small unfurnished rooms and kitchen at one end. Probably a 2 room hostel would be adequate on Cornwell's land – and would avoid the need for visitors to trek up to the site, 40 minutes in the early morning to see the adults flying – mostly 7.30 – 9.30 a.m.

We also considered the possibility of trying actively to encourage the development of eco-tourism on the plateau – initially based at Dareki, but decided against it for the following reasons:

(a) Severe access problems given the present state of the road.

(b) None of us have eco-tourism experience.

(c) There is a risk of unscrupulous black market dealers coming in.

It seems best if any genuine enthusiasts want to visit that they should make arrangements with Partners with Melanesians, who through Kenn Mondiai appear to be well liked in the area.

On the NBPOL front, matters appear to be in order on the ground, with a team from SDF visiting in early October to satisfy themselves on the financing and payment arrangements.

At a higher level the Governor of Oro Province, Gary Juffa, has expressed concerns, which have now been allayed by making him in his capacity as Governor Patron of the project.

Acknowledgements.

Charles Harbottle and I would like to thank most warmly David Mather, his staff and Vijou Vergis and his staff at Higaturu Estate. His staff accompanied us for 3 nights at Managalas, and did everything possible to make us comfortable under the somewhat spartan conditions of PNG village life. We are most grateful to them all.

26 September 2017

Appendix

Dear Mark,

Iam Kenn Mondiai, the Executive Director of Partners With Melanesians Inc., the national conservation and community development NGO working on the Managalas plateau in Oro Province, I am not happy with this press statement and it does not reflect on the work of PWM and our donor the Rainforest Foundation of Norway in Oro for the past 33 years
and the work we are doing with the local CBOs and the Umbrella Organisation for the people of Managalas plateau to protect the QABB. This new project looks like a one sided work to promote the NBPOL, Sime Darby Foundation just to fulfil your CSR Obligations, where the efforts of PwM and the local people is not at all acknowledged and what is there for the local people and there is no MOU to bridge the work between PWM, MCF and the local people …all is just talks and I don't welcome this Press Statement. STOP and fix the issues like an MOU before it goes out.

If this Press Statement Goes out as it is "We will not allow this project to work in Managalas plateau, and we will wash our hands from this work as there is no partnership"

Kenn Mondiai

Note It has been agreed that no further public announcements will be made until the NBPOL team and all others concerned have had a chance to consider the matter in further details.